

CO1-AT3(Computer Vision)

Technical Analysis of the Vivo Y18 Smartphone Camera: Hardware, Image Processing Pipeline and Artificial Intelligence Integration

1. Introduction

Smartphone cameras have evolved into advanced imaging systems by combining high-quality hardware with Artificial Intelligence (AI). The Vivo Y18 uses computational photography techniques and an integrated Image Signal Processor (ISP) to improve image quality despite being a budget smartphone.

The objective of this report is to analyze the Vivo Y18 camera system, explain the engineering techniques used to capture images, and understand how AI enhances photography.

2. Technical Specifications

Primary Camera Specifications

Specification	Details
Phone Model	Vivo Y18
Main Camera	50 MP Wide Camera
Secondary Camera	0.08 MP Auxiliary Lens
Front Camera	8 MP
Sensor Resolution	50 Megapixels
Sensor Size	Approximately 1/2.76-inch
Aperture	f/1.8
Focal Length	Approximately 26 mm (35 mm equivalent)
Pixel Size	0.64 μm
Pixel Binning	4-in-1 Pixel Binning



Specification	Details
Color Filter Array	Quad Bayer
Autofocus	Phase Detection Auto Focus (PDAF)
Optical Image Stabilization	Not Available
Electronic Image Stabilization	Available during video recording
Maximum Image Resolution	8192 × 6144 pixels
Video Recording	1080p @30fps

Camera Hardware Overview

Vivo Y18 Camera
50 MP Main Camera f/1.8
0.08 MP Depth Sensor
LED Flash

3. Sensor Details

Camera Sensor

The Vivo Y18 features a 50 MP CMOS image sensor. The CMOS sensor converts incoming light into electrical signals.

Sensor Characteristics

Feature	Description
Sensor Type	CMOS
Resolution	50 MP
Physical Size	1/2.76 inch
Pixel Size	0.64 μm



Feature	Description
Color Filter	Quad Bayer

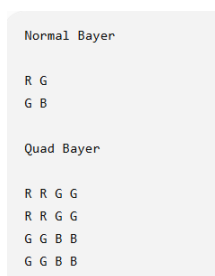
Working

Each pixel captures light.

The Bayer filter allows only one color (Red, Green or Blue) to each pixel.

The ISP later reconstructs the full-color image.

Pixel Architecture



Quad Bayer allows neighboring pixels to combine into one larger pixel for brighter images.

4. Optics

Parameter	Value
Aperture	f/1.8
Lens Type	Wide Angle
Equivalent Focal Length	26 mm
Lens Elements	5P Plastic Lens

Aperture

Large aperture allows:

- More light
- Better night photography
- Less noise

Stabilization

Type	Available
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Optical Image Stabilization (OIS)	No
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Electronic Image Stabilization (EIS)	Yes (Video)
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Since Vivo Y18 is a budget smartphone, it does not contain OIS.

Instead, it uses software-based stabilization.

Autofocus

The Vivo Y18 uses

Phase Detection Auto Focus (PDAF)

Advantages

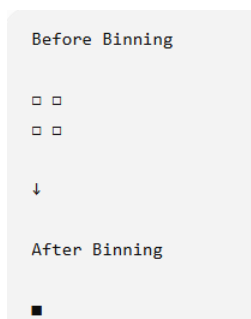
- Fast focusing
- Better object tracking
- Suitable for moving subjects

5. Implementation Techniques

Pixel Binning

The Vivo Y18 uses

4-in-1 Pixel Binning



Benefits

- Better low-light photography
- Reduced image noise
- Increased brightness

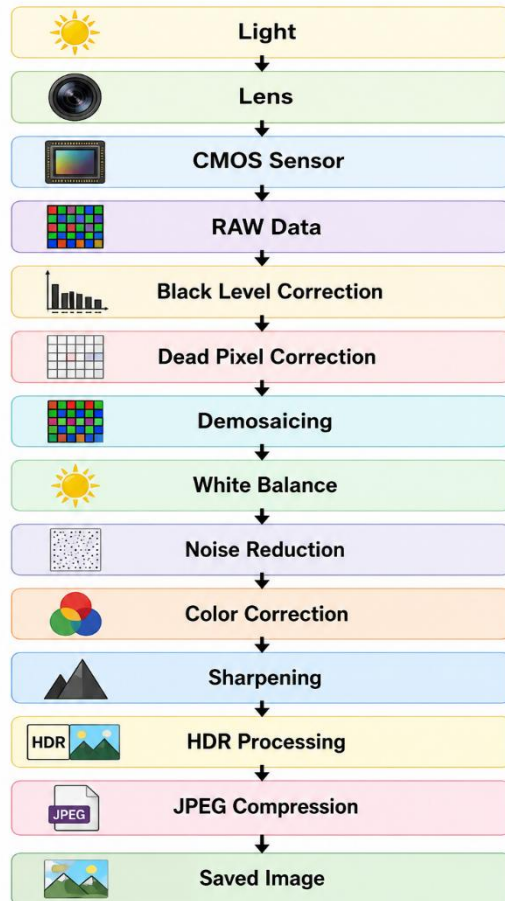
Effective output becomes approximately 12.5 MP.

Four neighboring pixels combine into one larger pixel.

Image Signal Processor (ISP)

The ISP is responsible for converting RAW sensor data into JPEG images.

ISP Pipeline



Sensor Shift vs Lens Shift

Lens Shift

Phone Shake



Lens Moves



Sensor Fixed

Used in premium smartphones.

Sensor Shift

Phone Shake



Sensor Moves



Lens Fixed

Used in flagship iPhones.

Vivo Y18

Neither Sensor Shift nor Lens Shift is available.

Instead it uses

Electronic Image Stabilization (EIS)

6. Artificial Intelligence Applications

AI Scene Recognition

The AI identifies

- Sky
- Trees
- Human face
- Food
- Pets
- Documents
- Flowers

Then automatically adjusts

- Saturation
- Brightness
- Contrast
- Sharpness

Semantic Segmentation

The camera divides the image into regions.

Example

Image

↓

AI Model

↓

Face

↓

Sky

Benefits

- Better sky color
- Better skin tone
- Better greenery
- Better portrait

↓

Grass

↓

Building

↓

Separate Optimization

HDR Computational Photography

Instead of taking one photo,

the Vivo Y18 captures

Dark Exposure

+

Normal Exposure

+

Bright Exposure

↓

AI Alignment

↓

HDR Image

Night Mode

Night Mode captures several images.

Frame 1

+

Frame 2

+

Benefits

- Better highlights
- Better shadows
- Higher dynamic range

Frame 3

+

Frame 4

+

Frame 5

↓

Image Alignment

↓

Noise Reduction

↓

Bright Night Image

Portrait AI

The AI detects

- Hair
- Face
- Background

Then

- Blurs background
- Preserves face details

Produces DSLR-like portraits.

Face Beauty AI

The camera automatically performs

- Skin smoothing
- Eye enhancement

- Face lighting
- Tone correction

AI Noise Reduction

Machine learning models remove

- Grain
- Sensor noise

while preserving

- Edges
- Texture

AI Color Enhancement

AI adjusts

- White Balance
- Saturation
- Exposure
- Dynamic Range

according to the detected scene.

Object Detection

The camera identifies

- Humans
- Food
- Pets
- Vehicles
- Flowers

Each object receives different optimization.

AI Camera Workflow

Open Camera



Scene Detection



Object Recognition



Semantic Segmentation



Exposure Optimization



HDR Processing



Noise Reduction



Face Enhancement



JPEG Image

7. Side-by-Side AI Comparison

AI OFF

AI ON

Normal Brightness Better Brightness

Flat Colors

Rich Colors

More Noise

Reduced Noise

AI OFF**AI ON**

Less Detail

Sharper Detail

Normal HDR

Enhanced HDR

8. Advantages

- ✓ High 50 MP resolution
- ✓ Quad Bayer technology
- ✓ Fast PDAF autofocus
- ✓ AI scene recognition
- ✓ HDR photography
- ✓ Night mode
- ✓ Portrait mode

9. Limitations

- ✗ No Optical Image Stabilization
- ✗ Small sensor compared to flagship phones
- ✗ Average low-light performance

10. Conclusion

The Vivo Y18 demonstrates how modern smartphones combine hardware and software to produce high-quality images. Its 50 MP Quad Bayer sensor, PDAF autofocus, and AI-powered computational photography enable impressive results for a budget device. Although it lacks premium features such as Optical Image Stabilization (OIS) and sensor-shift stabilization, the built-in ISP and AI algorithms compensate through HDR processing, noise reduction, scene recognition, semantic segmentation, portrait optimization, and night mode. This integration of imaging hardware, edge computing, and machine learning makes the Vivo Y18 capable of delivering clear, vibrant, and detailed photographs suitable for everyday use.